Caspian School of Academics

- Introduction
- Real Dataset Scenario
- Coding
  - pandas
  - Matplotlib
- Exercises
- Resources

Scan to ask

Turning Data into Insight

Data Analysis with pandas and Matplotlib

MICRO-TEACHING SESSION

Lecturer in Computing | Interview Presentation

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Caspian School of Academics

Today's Agenda

Introduction

Real Dataset Scenario

Coding

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Before We Start

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Outcome

By the end of this session, you will be able to:

Load and explore with pandas → **Visualise and interpret with Matplotlib** → **Complete a similar project**

Questions

Ask at any time:

Raise your hand

Scan the QR code

Why Is Data Valuable?

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WE HEAR IT ALL THE TIME

Data is valuable.

Useful information can be drawn from it.



But what does that actually mean?

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Data Is Everywhere

Every action we take can be recorded as data



But on its own, data has no value.

From Raw Data to Decisions

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Example: how a navigation app chooses your route

RAW DATA

```
52.37, 4.89  08:01  12
52.38, 4.90  08:02   9
52.40, 4.93  08:05  31
52.41, 4.95  08:09   8
52.43, 4.99  08:14  27
52.44, 5.02  08:20  14
52.47, 5.05  08:31   6
...
```

Data science
clean, analyse, visualise

INSIGHT



Route B is the fastest

Decision: take Route B and arrive 11 minutes earlier

Scenario: 50 Startups

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A dataset of 50 startups. You are the data analyst.

FEATURES (columns)

Sample

R&D Spend	Administration	Marketing Spend	State	Profit
165,349.20	**136,897.80**	**471,784.10**	**New York**	**192,261.83**
162,597.70	151,377.59	443,898.53	California	191,792.06
153,441.51	101,145.55	407,934.54	Florida	191,050.39
144,372.41	118,671.85	383,199.62	New York	182,901.99
…	…	…	…	…

50 startups = 50 samples, each described by 5 features

Coding

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Let's start coding

Open the notebook and code along

pandas → **Matplotlib**

Resources for Further Learning

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EDITORS

Jupyter Notebook
jupyter.org

VS Code
code.visualstudio.com

Google Colab
colab.research.google.com

LIBRARIES

pandas
pandas.pydata.org

Matplotlib
matplotlib.org

DATASETS

Kaggle
kaggle.com/datasets

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GitHub Links and Learning Videos

GITHUB

Today's lesson	github.com/zahrabakhshandeh/data-to-insight/
Practice exercises	github.com/zahrabakhshandeh/data-to-insight/tree/main/practice
Regression model (next step)	github.com/zahrabakhshandeh/linear-regression-from-scratch

LEARNING VIDEOS

Installing Jupyter	*youtube.com/your-video-link (sample link)*
Working with Google Colab	*youtube.com/your-video-link (sample link)*
Git and GitHub	*youtube.com/your-video-link (sample link)*

Note: for my students, I always record YouTube videos on learning the tools they need, such as Git, GitHub and Linux, and share the links. These videos are not in English, so I left them out of the main materials. If I start teaching in English, I will prepare them in English and share the links.

Contact

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Thank you!

Questions are welcome

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